

🎓 Education

2024-2026	M. Sc. in Information Processing Science - Faculty of Information Technology and Electrical Engineering, University of Oulu - CGPA: 4.88 / 5.00 Orientation: Information Systems Specialization: Human-Centered Design & User Experience, Persuasive Design & Gamification Thesis: <i>Making Friendship with Data Centers: Understanding Community Needs for Sustainability</i>, Grade: 5 / 5, Supervisor: Dr. Marianne Kinnula Reviewer: Dr. Dorina Rajanen, Dr. Marianne Kinnula, Dr. Netta Iivari	Oulu, Finland
2017-2022	B. Sc. in Computer Science & Engineering - Department of Computer Science & Engineering, Daffodil International University - CGPA: 3.92 / 4.00	Dhaka, Bangladesh

🔍 Research Experience

Masters Thesis

IT Infrastructure Sustainability

- Examined critical IT infrastructure (data center) operations for community sustainability through a qualitative study using the Sustainable HCI lens.
- Collected semi-structured, participatory design inputs from Oulu city residents and applied Thematic analysis to the data.
- Developed a human-centric data center operation and establishment guidelines for community sustainability.
- Connected data center operation with HCI and SHCI literature.
- Research Output:** [OuluRepo Thesis Link](#) | [Md Shafiul Alam](#), Marianne Kinnula, Dorina Rajanen. **Greenwashing is Cheap, Show Me the Numbers: Exploring Community Sustainability Requirements for Data Center Operation.** [Under Review].

Independent Work

CS students' and AI Sustainability

- Collaboration with researchers from Bangladesh on AI sustainability in a low-resource environment.
- Conducted an exploratory search on undergraduate Computer Science students' understanding of AI sustainability.
- Formulated curriculum update recommendations based on students' practices and limitations for disseminating the AI sustainability concept.
- Research Output:** Syed Tangim Pasha*, [Md Shafiul Alam](#)*, Md. Saad Bin Kamal*, Nusrat Jahan Nishat, Monjurul Ahsan Bhuiyan, Jannatun Noor. **Understanding Computer Science Students' Knowledge and Perceptions of AI Sustainability in Bangladesh.** [Under Review.] | [Pre-print](#).

* = first author

🔧 Skills

User Research	Participatory Design, Interviews, Think-aloud, Observation, Surveys, Thematic Analysis
UI Design	Figma, Low-fidelity Prototype, High-Fidelity Prototype, Wireframing
Programming	C, Java, Python, Assembly, JavaScript, PHP, HTML, CSS, XML, SQL, UML
Development	Unity, Visual Studio, Codeblocks, NetBeans, Android Studio
Library	NumPy, Pandas, Matplotlib, Scikit-Learn, OpenGL
Tools	GitHub, Zotero, L ^A T _E X, IBM SPSS, Mplus, XAMPP, Tinkercad, Emu8086, Google Colab, Cisco Packet Tracer
Hardware	Arduino
Language	Bengali (first language), English: C1 [IELTS: 4th Dec, 2023], Finnish: A1.3 [10 ECTS, University of Oulu course], Hindi: only listening and speaking

Notable Academic Projects

Details and Documentation of all the work can be found in the [Portfolio](#): 

Design and User Studies

- **Ecoland: Econews:** Designed, developed, and documented a climate change and News annotation sustainability game. | [GitHub](#) | [Project Paper](#)
- **XR Meditation:** Usability, user experience, and research data collection of a VR game. | [Google Drive](#)
- **DigiPet:** User experience design and user study of a retro game. | [Google Drive](#)
- **Securo Online:** Behavior change support system with a persuasive system framework for online privacy and security behavior change. | [Google Drive](#)
- **Multi-National Interact:** Design and User study of an immigrant community app. | [Google Drive](#)
- **Residente Attivo:** Design of Climate Change, Healthy Lifestyle, and Sustainability mobile game for urban population. | [Google Drive](#)

Development Works

- **Real-Time Password-Protected System:** Arduino based password-protected system. | [GitHub](#)
- **Eye's Suhrid:** An Eye well-being website. | [GitHub](#)
- **Onubadok:** Road sign translator Android app. | [GitHub](#)
- **Food Value Generator:** Assembly language based calorie calculator app. | [GitHub](#)

Teaching Experience

Fall 20 – Spring 21 **Lab Prefect**, CSE 232: Microprocessor, Embedded Systems & IOT Lab, Daffodil International University. (120 students)

Awards & Achievement

- **University of Oulu Scholarship** – M.S. Tuition Fee Waiver (2024-2026)
- **Tuition Fee Scholarship** – B.S. Tuition Fee Waiver, Daffodil International University (2017-2022)
- **IT Quiz Competition** – 1st Position, Daffodil ICT Carnival (2018)

Volunteer Contribution

- Information Processing Science thesis template updated to the latest version in Overleaf. (2026) | [Template Link](#)

Undergraduate Students

- **Ifat Bin Amin** — B.S. in Electrical and Electronics Engineering, North South University, Dhaka, Bangladesh, 2025-Present